Week 11: Cluster Monitoring - Lab Guide

HPC & Cluster Computing

Objective: Deploy a complete monitoring stack. **Goal**: Visualize the real-time CPU and RAM usage of your cluster.

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1. Prerequisites

Note: This lab assumes your Virtual Cluster (configured in previous weeks) is **up and running**. We are adding services to the existing live system.

- [ ] **Master Node**: Will host Prometheus and Grafana.
- [ ] **Compute Nodes**: Will host Node Exporter.
- [ ] **Networking**: Port 9090 (Prometheus) and 3000 (Grafana) must be open.

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2. Deploy Node Exporter (Compute Nodes)

We need to start generating metrics.

2.1 Update Warewulf Image

Since our nodes are stateless, we add the exporter to the container.

- [ ] Enter the container: `bash sudo wwctl container exec rocky-9 /bin/bash`
- [ ] Download and Install (Inside Container): ```bash # Download wget https://github.com/prometheus/node_exporter/releases/download/v1.7.0/node_exporter-1.7.0.linux-amd64.tar.gz tar xvf node_exporter-*.tar.gz cp node_exporter*-.linux-amd64/node_exporter /usr/local/bin/

Create Systemd Service

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cat < /etc/systemd/system/node_exporter.service [Unit] Description=Node Exporter After=network.target

[Service] User=root ExecStart=/usr/local/bin/node_exporter

[Install] WantedBy=multi-user.target EOF

Enable

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systemctl enable node_exporter - `[ ] Exit and Rebuild:`bash exit sudo wwctl container build rocky-9 sudo wwctl power cycle node[01-04] ```

2.2 Verify

- [ ] Check if a node is reporting: `bash curl http://node01:9100/metrics | head` *Did you see lines starting with* `node_`*?*

> 📝 ***Question 1**: Look at the output of the curl command. What is the value of `node_context_switches_total`? What does this number imply about the OS kernel?*

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3. Deploy Prometheus (Master)

3.1 Install

- [ ] Download and Install: `bash wget https://github.com/prometheus/prometheus/releases/download/v2.45.0/prometheus-2.45.0.linux-amd64.tar.gz tar xvf prometheus-*.tar.gz sudo mv prometheus-*.linux-amd64 /opt/prometheus`

3.2 Configure Targets

- [ ] Edit `/opt/prometheus/prometheus.yml`: ```yaml scrape_configs:
  - job_name: 'cluster' static_configs:
    - targets: ['node01:9100', 'node02:9100', 'node03:9100', 'node04:9100'] ```

> 📝 ***Question 2**: Why do we specify port `9100` for the targets, but access Prometheus itself on port `9090`?*

3.3 Start Service

- [ ] Run Prometheus (Background): `bash /opt/prometheus/prometheus --config.file=/opt/prometheus/prometheus.yml &`
- [ ] **Verification**: Open Browser to `http://<master-ip>:9090/targets`. Are all endpoints **UP**?

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4. Deploy Grafana (Master)

4.1 Install

- [ ] Install RPM: `bash sudo dnf install -y https://dl.grafana.com/oss/release/grafana-10.0.0-1.x86_64.rpm`
- [ ] Start Service: `bash sudo systemctl enable --now grafana-server`

4.2 Login

- [ ] Open Browser: `http://<master-ip>:3000`.
- [ ] Default Login: `admin` / `admin`.

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5. Build the Dashboard (Interactive)

5.1 Add Data Source

- [ ] Click **Connections** -> **Data Sources** -> **Add new**.
- [ ] Select **Prometheus**.
- [ ] URL: `http://localhost:9090`.
- [ ] Click **Save & Test**. *Did it say "Successfully queried"?*

5.2 Create Dashboard

- [ ] Click **Dashboards** -> **New**.
- [ ] **Add Visualization**.
- [ ] Select **Prometheus** as source.
- [ ] Query: `node_memory_MemTotal_bytes`.
- [ ] Click **Run Queries**. *Result*: You should see a graph of memory usage for all your nodes!

> 📝 ***Question 3**: Run a stress test on `node01` (e.g., `stress --cpu 4`). Watch the Grafana dashboard. How long is the delay between starting the stress test and seeing the spike on the graph? Why is there a delay?*

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6. Submission

Create a **Lab Report PDF** containing:

1. **Answers**: Provide your written answers for Questions 1, 2, and 3.
2. **Screenshots**:
    - Prometheus Targets page showing all nodes UP.
    - Your Grafana Dashboard showing live data (with the stress test spike visible).

Course Index